1. Solve $x/3 + 1/5 = x/2 - 1/4$

2. Show that $x = 4$ is a solution of the equation $x + 7 - 8x/3 = 17/6 - 5x/8$.

3. Find x for the equation: $(2 + x)(7 - x)/(5 - x)(4 + x) = 1$

4. A number is such that it is as much greater than 45 as it is less than 75. Find the number.

5. Divide 40 into two parts such that 1/4th of one part is 3/8th of the other.

The digits of a 2-digit number differ by 5. If the digits are interchanged and the resulting number is added to the original number, we get 99. Find the original number.

What should be subtracted from thrice the rational number -8/3 to get 5/2?

The sum of three consecutive multiples of 7 is 63. Find these multiples.

The numerator of a fraction is 2 less than the denominator. If one is added to its denominator, it becomes 1/2 find the fraction.